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REVIEW ARTICLE

The Influence of Post-Exercise Cold-Water Immersion on Adaptive Responses to Exercise: A Review of the Literature

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Abstract Post-exercise cold-water immersion (CWI) is used extensively in exercise training as a means to minimise fatigue and expedite recovery between sessions. However, debate exists around its merit in long-term training regimens. While an improvement in recovery following a single session of exercise may improve subsequent training quality and stimulus, reports have emerged suggesting CWI may attenuate long-term adaptations to exercise training. Recent developments in the understanding of the molecular mechanisms governing the adaptive response to exercise in human skeletal muscle have provided potential mechanistic insight into the effects of CWI on training adaptations. Preliminary evidence suggests that CWI may blunt resistance signalling pathways following a single exercise session, as well as attenuate key long-term resistance training adaptations such as strength and muscle mass. Conversely, CWI may augment endurance signalling pathways and the expression of genes key to mitochondrial biogenesis following a single endurance exercise session, but have little to no effect on the content of proteins key to mitochondrial biogenesis following long-term endurance training. This review explores current evidence regarding the underlying molecular mechanisms by which CWI may alter cellular signalling and the long-term adaptive response to exercise in human skeletal muscle.

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Key Points

Despite short-term increases in markers of mitochondrial biogenesis following a single session of post-exercise cold-water immersion (CWI), regular post-exercise CWI appears to have little to no effect on long-term endurance training adaptations in human skeletal muscle.

Evidence to date suggests either no effect or negative effects of post-exercise CWI on the molecular mechanisms regulating resistance training adaptations, as well as key resistance training adaptations (e.g., strength and muscle mass).

Research supporting or refuting the use of post-exercise CWI during endurance or resistance training programmes is scarce, and further research is needed to determine its influence on human skeletal muscle adaptations during real-world training scenarios.

1 Introduction

Recovery from exercise refers to the restoration of the body's physiological and psychological processes to a pre-fatigue state and performance level [1]. A large body of research has focused on modalities designed to hasten recovery, with one of the most prevalent techniques being cold-water immersion (CWI). However, given the widespread use of CWI as a recovery modality, as well as the contrasting findings reported to date, clarification of the

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effects of CWI on the skeletal muscle adaptations to exercise training in humans is warranted. This review summarises the current evidence that post-exercise CWI alters the adaptive response to exercise in human skeletal muscle, with particular focus on the regulation of mitochondrial biogenesis and muscle protein turnover.

2 Literature Search

A literature search was performed on SCOPUS® with the following keywords: 'cold', 'skeletal muscle' and 'exercise'. From the abstracts returned (1963 to December 2017), articles were included in the review if they related to the molecular mechanisms by which cold, administered during or immediately after an exercise session, may alter skeletal muscle cellular responses to exercise. Articles were reviewed if they related to intracellular signalling, gene transcription, and protein translation associated with mitochondrial biogenesis and muscle protein turnover. Reference lists of the included articles were also searched and additional articles satisfying the above criteria were included for review.

3 Cold-Water Immersion (CWI)

Cold-water immersion has emerged as a popular cryotherapy tool aimed at enhancing recovery following training and competition [2, 3]. Typically utilised following exercise, it involves the immersion of all or part of the body in cold water, with an intended therapeutic outcome. Although an optimal prescription for post-exercise CWI does not currently exist, water temperatures cooler than 15 °C and immersions of at least 10 min are typically utilised [1].

Post-exercise CWI is believed to limit, and speed the recovery of, exercise-induced decrements in functional capacity and exercise performance [1, 2]. For a detailed review of the effects of CWI on the recovery from exercise, the reader is referred to a review on this topic [1]. Briefly, many studies have demonstrated beneficial improvements in muscle strength [4–6], aerobic exercise performance [7–9], markers of exercise-induced muscle damage [4, 10, 11], inflammation [3, 12–14], muscle soreness [5, 6, 15–17] and perceptions of fatigue [12, 16–18] when CWI was implemented as a recovery technique. However, CWI has also been demonstrated to have no influence [5, 11–14, 19–29], or even a detrimental effect [18, 30–34] on similar measures, highlighting the equivocal nature of CWI research. The lack of consistency in the observed effects of CWI is commonly attributed to methodological discrepancies between studies, such as the immersion and/or exercise protocols used, as well as the performance outcomes measured [15]. Another explanation for these discrepancies is a placebo effect related to belief in the effectiveness of CWI [35].

Although the specific mechanisms by which CWI might hasten recovery remain to be fully elucidated, increased hydrostatic pressure and/or a reduction in muscle temperature may play important roles [2]. Hydrostatic pressure will cause an upward and inward displacement of body fluid, aiding the return of fluid from the muscles to the blood [2]. A resultant reduction in muscle oedema might improve substrate transportation, the metabolism of waste products [20, 36], and the maintenance of muscle function and repair [2]. A reduction in tissue temperature is believed to promote many physiological responses, including analgesia, reduced metabolism and enzyme activity, and a vasoconstrictive response to reduce blood flow to the muscle [37]. As a result, CWI may diminish the effects of soft tissue injury [38] via an attenuation of oedema, inflammation, pain, and haematoma formation [37].

It has been suggested the physiological alterations associated with post-exercise CWI may serve to improve subsequent training load and quality [39, 40], potentially providing a greater stimulus for adaptation. However, conclusive evidence regarding the merit of CWI in improving training quality is lacking, with research both supporting [41] and refuting [31] these suggestions. A potential limitation with assessing this hypothesis is the difficulty in quantifying the effects of post-exercise CWI on subsequent training quality, which may be influenced by a number of factors (e.g., nutrition, motivation and training history) [39]. Furthermore, the markers commonly assessed (e.g., strength and endurance performance) provide little mechanistic insight on the effects of regular CWI on the adaptive response in skeletal muscle. Recent studies investigating the effects of CWI on adaptive molecular markers in skeletal muscle have begun to fill this gap.

4 Potential Influence of CWI on the Adaptive Responses to a Single Exercise Session

Skeletal muscle is a highly malleable tissue, capable of altering muscle protein abundance in response to disruptions in cellular homeostasis. Exercise provides one such disruption, and accumulative disruptions to cellular homeostasis via repeated sessions of exercise, termed training, can drastically alter muscle phenotype. Exercise can broadly be grouped into aerobic/endurance or strength/resistance, with the stimulus intensity, duration and frequency determining numerous morphological and metabolic muscular adaptations [42]. Endurance exercise is predominantly associated with skeletal muscle phenotype

alterations such as increased mitochondrial content, mitochondrial function and aerobic power [42–44], whereas resistance exercise is more closely associated with increased muscle mass, fibre hypertrophy and strength [42–44].

Many stimuli are implicated in the activation of cellular cascades that lead to muscle adaptations following exercise, including mechanical strain, ATP turnover, calcium flux, redox balance, reactive oxygen species (ROS) production, and intracellular oxygen pressure [44, 45]. These stimuli serve to activate signalling kinases and phosphatases, which in turn generate and/or activate transcription factors and co-regulator proteins that regulate target genes [45]. Depending on the initial stimulus, these complex networks exert molecular control over contractile, metabolic and mitochondrial adaptations [44].

4.1 Molecular Responses Affected by Cold

Ultimately acting as a stress signal, variations in temperature influence a wide array of cellular responses in skeletal muscle. Cold conditions can induce (1) alterations to messenger RNA (mRNA) stability, transcription and translation, (2) denaturation and a generalised reduction in protein synthesis, (3) decreased metabolic rate, (4) initiation of cell cycle arrest/apoptotic pathways, and (5) disruptions to cell membrane permeability and cellular cytoskeletal structure [46–50]. By influencing these processes, temperature may affect certain adaptive mechanisms, including the regulation of exercise-induced mitochondrial biogenesis [51] and muscle protein turnover [52, 53] (Fig. 1).

An innate ability to survive and adapt to cold stress is dependent on the conserved and co-ordinated induction of both cold- (CSP) and heat-shock proteins (HSP), whose primary function is to protect from and prepare for cellular stresses, including those imposed by thermal shock [54]. During cold exposure, CSP expression is rapidly and markedly increased, most notably cold-inducible RNA-binding protein (CIRP) and putative RNA-binding motif protein 3 (RBM3). In response to mild hypothermia (maximal expression ~ 32 °C), these proteins act as RNA chaperones to enhance translation via mRNA stabilisation or act to facilitate global protein synthesis [48, 55]. Following the initial cold shock, gene expression is also altered upon rewarming [50, 56, 57], including changes typically observed following heat stress (e.g., HSP expression). The main function of HSPs is to act as protein chaperones, by which they assist the refolding of denatured proteins and stabilise the confirmation of proteins important for a variety of cellular regulatory processes [48, 58, 59]. Further to this, HSPs regulate cellular redox state and signal transduction [60], protein turnover [58], and inflammation [61], as well as having an important role in the transportation of proteins involved in mitochondrial biogenesis [62]. As such, and via the up-regulation of CSPs and HSPs, cold stress has the ability to alter a variety of cellular responses to exercise, including intracellular signalling, gene transcription and protein translation [50].

Exposure to cold may also activate thermogenesis, the process whereby energy expenditure and metabolic demand are increased in an attempt to produce heat. For example, repeated muscular contractions during shivering thermogenesis may increase resting metabolic rate by ~ five-fold [63] and oxygen consumption by ~ two-fold [51, 64], by which a natural by-product is heat (and contraction-induced increases in myoplasmic Ca^{2+} and AMP) [51]. Non-shivering thermogenesis is characterised by the recruitment of uncoupling proteins (UCPs) to produce heat instead of shivering [65, 66]. These proteins allow the dissipation of the proton gradient before it can be used to produce ATP, thus allowing chemical energy to be dissipated as kinetic energy to produce heat [66]. Cold exposure also causes the uncoupling of sarcoplasmic reticulum Ca^{2+}-ATPase (SERCA) via sarcolipin (a SERCA regulator), leading to an increase in myoplasmic Ca^{2+} and non-shivering thermogenesis [67]. Similar to shivering thermogenesis, non-shivering thermogenesis is accompanied by increases in AMP:ATP, oxygen consumption/metabolic rate, and mitochondrial-uncoupled respiration [68, 69]. Contraction- and uncoupling-induced increases in free cytosolic Ca^{2+} and AMP provide a means by which cold exposure may alter the adaptive response to both endurance [70, 71] and resistance [52, 72] exercise.

Cold exposure has also been implicated in the production of ROS. For example, both shivering and non-shivering thermogenesis elevate metabolic rate and stimulate mitochondrial respiration, of which a natural by-product is ROS [73]. ROS production may be further exacerbated during rewarming [56], and by the oxidation of catecholamines (increased following CWI) [74, 75]. Although high levels of ROS have been reported to induce DNA damage [76, 77], alter enzyme function [78–80], reduce protein turnover [81–83] and compromise cellular integrity [78, 84], low-to-moderate levels of ROS exert positive effects on the immune system and essential metabolic functions [85]. For example, ROS play regulatory roles in cellular signalling pathways, gene expression, and skeletal muscle force generation [86–89]. Given their apparent role in regulating cellular responses, elevations in ROS may provide an alternate explanation by which post-exercise CWI can alter the adaptive response to exercise.

Another potential mechanism by which cold-induced alterations in ROS may influence cell signalling is via the tumour suppressor protein p53. Recent research has highlighted a complex interaction between ROS and p53 [146],

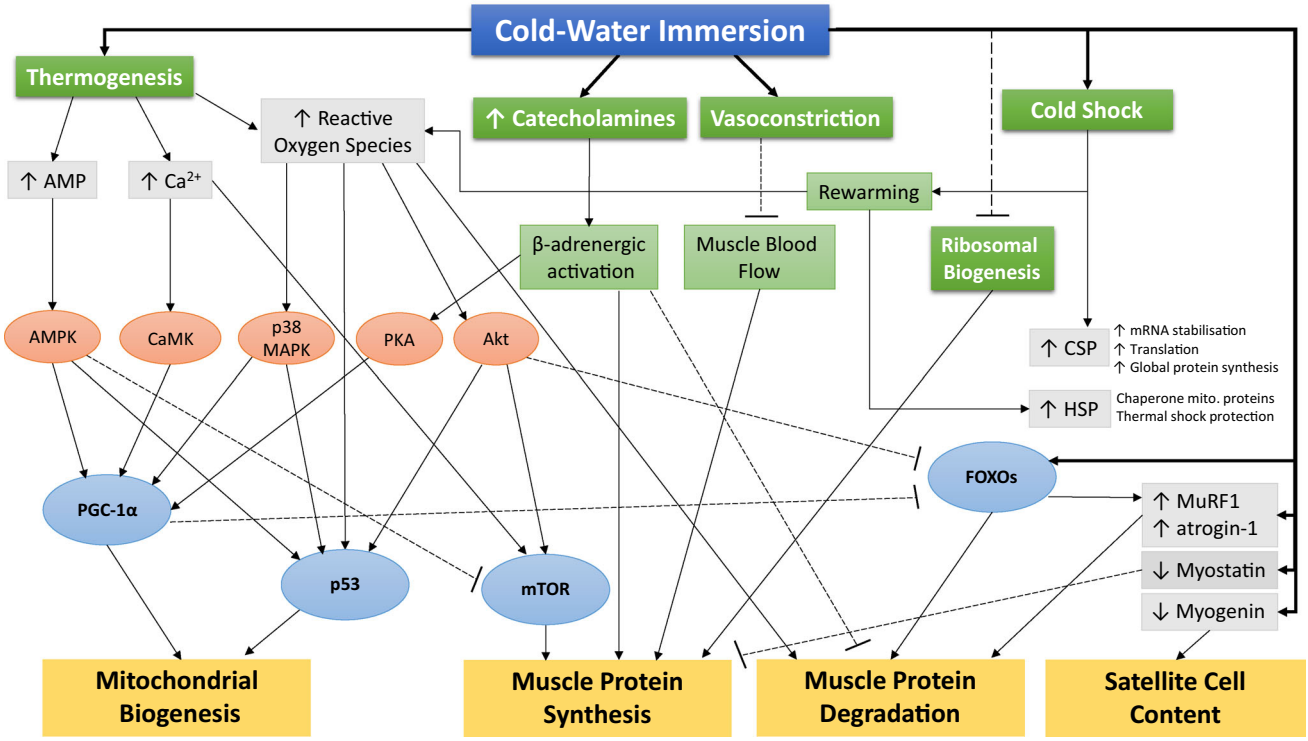

Fig. 1 Potential molecular mechanisms by which post-exercise cold-water immersion may alter the adaptive response to exercise. *Akt* protein kinase B, *AMP* adenosine monophosphate, *AMPK* adenosine monophosphate-dependent protein kinase, *Ca^{2+}* calcium, *CaMK* calcium/calmodulin-dependent kinase, *CSP* cold-shock proteins, *FOXOs* forkhead box class O, *HSP* heat-shock proteins, *MAPK* mitogen-activated protein kinase, *mTOR* mechanistic target of rapamycin, *MuRF1* muscle ring-finger protein-1, *p53* tumour suppressor protein p53, *PGC-1α* peroxisome proliferator-activated receptor gamma coactivator 1-α, *PKA* protein kinase A, → increase/activate; —| reduce/inhibit

in which ROS may both trigger p53 activation and be modulated by p53 [146]. p53 has also been shown to be activated following thermal shock [147, 148], with suggestions that hypothermic and subsequent rewarming shocks may induce a DNA-damage response [62]. A well accepted role of p53 in response to cold-shock is the regulation of cell cycle arrest, evident following mild [147, 149] and severe [150] hypothermia in rodents and cultured cells.

Cold exposure may also alter the molecular response to exercise via sympathetic nervous system and adrenergic receptor activation. For example, cold has been shown to increase noradrenaline concentrations [90], which in turn may stimulate the β_3-adrenergic receptor and the downstream protein kinase A (PKA)/cAMP pathway [91, 92], and subsequently mitochondrial biogenesis. Cold exposure may also increase muscle mass, as administration of β-adrenergic agonists has also been reported to promote muscle growth, potentially via an increase in muscle protein synthesis (MPS) and/or a decrease in muscle protein breakdown (MPB) [93, 94]. However, although there is independent evidence that cold exposure increases catecholamine concentrations [90], as well as evidence that catecholamines may activate mitochondrial biogenic pathways [91, 92] and regulate muscle mass [93, 94], evidence to support a correlational link between these processes is currently lacking.

4.2 Responses to Endurance Exercise

One of the most dramatic changes in skeletal muscle following endurance exercise is mitochondrial biogenesis [44, 45]. Exercise-induced mitochondrial biogenesis corresponds closely to cellular energy requirements and functions to improve oxidative phosphorylation, ATP yield, substrate utilisation and fatigue resistance [62, 95]. Central to the process of mitochondrial biogenesis, the regulatory proteins peroxisome proliferator-activated receptor gamma (PPARγ) coactivator 1-alpha (PGC-1α) and p53 play important roles in regulating mitochondrial content and respiratory function [91, 96–103]. For a more detailed description of the effects of endurance exercise on mitochondrial adaptations, and the role of key regulatory proteins in mitochondrial biogenesis, readers are referred to reviews on this topic [45, 104].

A summary of studies investigating the effects of cold exposure following endurance exercise on skeletal muscle responses is presented in Table 1. These altered responses may be largely related to thermogenesis and mediated by PGC-1α. For example, shivering-induced increases in Ca^{2+} and AMP may hypothetically increase PGC-1α mRNA via the activation of the calcium-sensitive calcineurin and calcium/calmodulin-dependent protein kinase (CaMK) [70, 105], and the energy sensitive AMP protein kinase (AMPK) [71, 106], respectively. In support of this, post-exercise cold exposure (3–4 h at 7 °C) in humans was shown to increase PGC-1α mRNA [51, 64], consistent with prior work in animal models [65, 107–109]. However, these increases in PGC-1α mRNA were not associated with changes in downstream markers of mitochondrial biogenesis, including cytochrome *c* oxidase (COX), UCP3, mitofusin-2 (Mfn2), nuclear respiratory factor-1 (NRF-1) or mitochondrial transcription factor A (Tfam) mRNA [51, 64]. Although these findings suggest a limited effect of shivering thermogenesis on mitochondrial biogenesis, it is unlikely that 3–4 h was long enough to accurately assess the effect of thermogenesis-induced increases in PGC-1α mRNA on the activation of processes downstream of PGC-1α.

Enhanced mitochondrial biogenesis is an important component of non-shivering thermogenesis, in which PGC-1α also plays an important mediatory role [96]. Cold exposure has been reported to increase PGC-1α mRNA, independent of shivering, in brown adipose tissue [91], rodent [110] and chicken [109] skeletal muscle, and more recently in human skeletal muscle following exercise [111–113]. Although largely unknown, this response may be mediated by the β_3-adrenergic and PKA/cAMP pathways [91, 92], altered Ca^{2+} handling and CaMK/calcineurin activation [65], and/or increased AMP/ATP ratio and AMPK activation [107] (Fig. 1). Given the well established role of PGC-1α in the regulation of mitochondrial biogenesis [114], cold-induced increases in PGC-1α mRNA provide a plausible molecular basis by which post-exercise CWI might enhance exercise-induced mitochondrial biogenesis [96].

Four studies have investigated the effects of post-exercise CWI on signalling responses associated with mitochondrial biogenesis in humans [111–113, 117] (Fig. 2). CWI administered after a single session of high-intensity running [111, 112] or cycling [113] increased PGC-1α mRNA content in a cooled lower limb (up to ~ 9.0-fold), which was larger than the response in the contra-lateral and non-cooled control limb (up to ~ 5.0-fold). Furthermore, these CWI-induced increases in PGC-1α mRNA may occur independent of a preceding exercise stimulus [112], and systemically [113]. More recently, a similar increase (~ 7.5-fold) in PGC-1α mRNA content was observed following a single session of sprint-interval cycling and post-exercise CWI [117]. This was, however, not significantly larger than the increase seen in the control condition (~ 5.4-fold), suggesting the effects of CWI on PGC-1α mRNA may be more pronounced following lower-intensity exercise. Despite the CWI-induced increases in PGC-1α mRNA reported in Fig. 2, CWI appears to have no effect on the mRNA content of other proteins associated with PGC-1α and mitochondrial adaptations (at least within 3 h post-exercise), including NRF-1/2, Tfam, UCP3, sirtuin-1, COX, citrate synthase (CS) and estrogen-related receptor α [113, 117]. Alternatively, it is possible that the 3-h-postrecovery biopsy time point may not have adequately captured the time course of changes in mRNA content of these proteins.

The potential mechanisms by which post-exercise CWI increases PGC-1α mRNA have been discussed recently. Given the apparent lack of shivering reported by Ihsan et al. [111], the increase in PGC-1α mRNA was attributed to cold-induced β-adrenergic activation, and a subsequent increase in nitric oxide synthesis. However, no differences in nitric oxide synthase markers were reported between the cold and control leg, suggesting the increase in PGC-1α mRNA may be explained by other upstream pathways (e.g., AMPK and CaMK pathways). Allan et al. [113] suggest the 'systemic' increase in skeletal muscle PGC-1α mRNA (i.e., both legs) following post-exercise CWI to a single leg may be the result of increased catecholamine levels and subsequent adrenergic activation of AMPK. Only two studies have investigated the effects of post-exercise CWI on $AMPK^{Thr172}$ phosphorylation, both reporting no effect immediately post-exercise when compared with a passive control [113, 117]. Cold exposure can also alter Ca^{2+} release [127] and re-uptake [128] within skeletal muscle not involved in shivering, in turn increasing free myoplasmic Ca^{2+} and potentially inducing CaMK-induced PGC-1α transcription [70]. To date, no data exist investigating the effects of post-exercise CWI on CaMK activation.

An alternate explanation for increased PGC-1α mRNA following post-exercise CWI is a cold-induced increase in ROS. Potentially serving as a compensation for altered energy demands and respiratory chain defects [129, 130], low-to-moderate levels of ROS may have a beneficial, hormetic effect on the mitochondria [131]. This response may be mediated by PGC-1α [130], with ROS accumulation reported to activate upstream kinases p38 MAPK [132] and protein kinase B (Akt) [133]. However, post-exercise CWI was shown to have no effect on p38 $MAPK^{Thr180/Tyr182}$ phosphorylation [113, 117], with data on Akt phosphorylation currently lacking. Nuclear factor-κB (NF-κB) is also a redox-sensitive transcription factor, which via ROS-induced activation of IkB-α kinase (IKK) [134] may

Table 1 A summary of current evidence regarding the short- and long-term effects of cold exposure following endurance exercise on skeletal muscle responses

Study	Sample size (sex)	Species	Exercise protocol	Cold intervention		Muscle alterations (vs control)	
				Single	Repeated	mRNA	Protein/other
Christiansen et al. [115]	19 (M)	Human	Cycling SIT 4 × 30-s 'all-out'	CWI ~ 10 °C for 15 min		↑ NKAα_2, Sp1 ↔ NKA$\alpha_{1,3}$, NKA$\beta_{1\text{-}3}$, FXYD1, HIF-1α, Sp3	
Christiansen et al. [116]	12 (M)	Human	Cycling SIT 3 day/week, 6 week		CWI ~ 10 °C for 15 min	–	↔ NKA$\alpha_{1\text{-}3}$, NKA$\beta_{1\text{-}3}$, FXYD1
Broatch et al. [117]	19 (M)	Human	Cycling SIT 4 × 30-s 'all-out'	CWI ~ 10 °C for 15 min		↔ NRF1/2, Tfam, PGC-1α, p53, MFN2, AIF, SCO2, DRP1, CIRP, RBM3, UCP3	↔ p-p53, p-p38 MAPK, p-AMPK
	16 (M)	Human	Cycling SIT 3 day/week, 6 week		CWI ~10 °C for 15 min	–	↔ PGC-1α, p53, Mito. respiration, CS activity
Shute et al. [118]	8 (M)	Human	Cycling 90 min, variable intensity	Local ice application 4 h after exercise		↔ NRF1/2, Tfam, PGC-1α, ERRα	–
Allan et al. [113]	10 (M)	Human	Cycling HIT (8 × 5 min at ~ 80% peak power output)	CWI ~ 8 °C for 10 min		↑ PGC-1α ↔ SIRT1, NRF2, p53, COX, CS, Tfam, ERRα	↑ p-AMPK ↔ p-p38 MAPK, PGC-1α
Joo et al. [112]	9 (M)	Human	Running HIT 8 × 3 min (90% VO_{2max})	CWI ~ 8 °C for 10 min		↑ PGC-1α, $VEGF_{total}$ ↔ $VEGF_{165}$	↔ PGC-1α, VEGF
	10 (M)	Human	– (Passive CWI)	CWI ~ 8 °C for 10 min		↑ PGC-1α, $VEGF_{165}$ ↔ $VEGF_{total}$	↔ PGC-1α, VEGF
Aguiar et al. [119]	17 (M)	Human	Cycling HIT (90–110% W_{max}) 3 day/week, 4 week		CWI ~10 °C for 15 min	↑ Tfam, CaMK2, PDK4, HSF ↔ NRF1/2, PGC-1α, CPT1, CS	↔ Hsp72, p-p38 MAPK, AMPK
Ihsan et al. [120]	8 (M)	Human	Running HIT (80–110% V_{max}) 3 day/week, 4 week		CWI ~10 °C for 15 min		↑ β-HAD, AMPK, p-AMPK, p-ACC, OXPHOS I and III ↔ CS, PGC-1α, PPARα, p38 MAPK, OXPHOS II, IV and V ↔ CS, CPT1 and SDH activity
Ihsan et al. [111]	9 (M)	Human	Running 30 min (70% V_{max}) + HIT to exhaustion (100% V_{max})	CWI ~ 10 °C for 15 min		↑ PGC-1α	–

Table 1 continued

Study	Sample size (sex)	Species	Exercise protocol	Cold intervention		Muscle alterations (vs control)	
				Single	Repeated	mRNA	Protein/other
Xing et al. [121]	100 (M + F)	Rats	Swimming training 30 min 7 day/week, 6 week		CWI 4 °C while training		↑ AC, PDE and NKA activity (during first 3 week)
Slivka et al. [64]	8 (M)	Human	Cycling 1 h (65% VO_{2max})	Cold air 7 °C during exercise and 4 h of recovery		↑ PGC-1α ↓ ERR-α, NRF-2	
Slivka et al. [51]	9 (M)	Human	Cycling 1 h (60% W_{max})	Cold air 7 °C during exercise and 3 h of recovery		↑ PGC-1α	
Camargo et al. [122]	32 (M)	Rats	Swimming 100 min	CWI 10 °C for 10 min			↓ Necrosis ↓ leukocyte infiltration
Watkins et al. [123]	8 (M)	Human	Cycling 2 × 30 min (75% VO_{2peak})	Cold air 9 °C during exercise			↔ HSP72
Schaeffer et al. [124]	48 (M)	Possum	Running training 30 min (10 $m.min^{-1}$) 5 day/week, 9 week		Cold air 9–19 °C for 9 week		↑ Muscle mitochondrial volume density
Young et al. [125]	18 (M)	Human	Cycling training 1 h (60% VO_{2max}) 5 day/week, 8 week		CWI 20 °C while training		↔ CS activity
Seebacher and Glanville [108]	36 (M + F)	Rats	Running training 30 min (60% W_{max}) 5 day/week, 30 d		Cold air 12 °C for 30 d	↑ NRF-1 and UCP1	–

Table 1 continued

Study	Sample size (sex)	Species	Exercise protocol	Cold intervention		Muscle alterations (vs control)	
				Single	Repeated	mRNA	Protein/other
Mitchell et al. [126]	32 (M)	Rats	Running training 60 min (20 m.min^{-1}) 5 day/week, 9 week		Cold air 8 °C and 4 °C for 9 week		↑ COX and mtHSP70

AC adenylyl cyclase, *ACC* acetyl-CoA carboxylase, *AIF* apoptosis-inducing factor, *AMPK* adenosine monophosphate protein kinase, *β-HAD* beta-hydroxyacyl dehydrogenase, *CaMK* calcium/calmodulin-dependent kinase, *CIRP* cold-inducible RNA-binding protein, *COX* cytochrome c oxidase, *CPT1* Carnitine palmitoyltransferase I, *CS* citrate synthase, *CWI* cold-water immersion, *DRP1* dynamin-related protein 1, *ERR-α* estrogen-related receptor alpha, *FXYD1* phospholemman isoform 1, *HIF-1α* hypoxia-inducible factor 1α, *HIT* high-intensity interval training, *HSF* heat-shock factor, *HSP72* heat-shock protein 72, *MAPK* mitogen-activated protein kinase, *MFN2* mitofusion-2, *mtHSP70* mitochondrial heat-shock protein 70, *NKA* Na^+/K^+ -ATPase isoform, *NRF* nuclear respiratory factor, *p53* tumour suppressor protein p53, *PDE* phosphodiesterase, *PDK4* mitochondrial pyruvate dehydrogenase lipoamide kinase isozyme 4, *PGC-1α* peroxisome proliferator-activated receptor gamma coactivator 1-alpha, *RBM3* putative RNA-binding protein 3, *SCO2* synthesis of cytochrome c oxidase 2, *SDH* succinate dehydrogenase, *SIRT1* sirtuin-1, *SIT* sprint-interval training, *Sp* specificity protein, *Tfam* mitochondrial transcription factor, *UCP* uncoupling-protein, *VEGF* vascular endothelial growth factor, $VO_{2max/peak}$ maximal/peak oxygen uptake, V_{max} maximal aerobic running velocity, W_{max} maximal aerobic power, ↑ significant ($p < 0.05$) increase, ↓ significant ($p < 0.05$) decrease, ↔ no effect ($p > 0.05$)

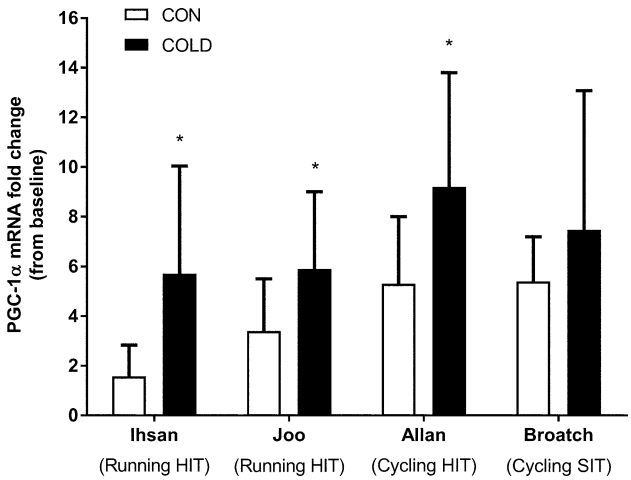

Fig. 2 Studies performed to date investigating the response of post-exercise cold-water immersion (*COLD*), as compared with passive control (*CON*), on peroxisome proliferator-activated receptor gamma coactivator 1-alpha (*PGC-1α*) mRNA content, 3 h after exercise. Data is from studies by Broatch et al. [117], Allan et al. [113], Joo et al. [112], Ihsan et al. [111]. *HIT* high-intensity interval training, *SIT* sprint-interval training. *Significantly different from CON

provide an alternate link between cold-induced ROS production and mitochondrial biogenesis [135]. Post-exercise CWI has also been reported to increase the mRNA content of the Na^+/K^+-ATPase (NKA) α_2 isoform, and redox-sensitive transcription factor specificity protein 3 (Sp3) [115]. Since NKA plays a key role in maintaining skeletal muscle membrane excitability during exercise [136], CWI may be a novel method by which to regulate NKA expression and subsequent skeletal muscle contractility/exercise performance [137]. Despite evidence supporting the notion that CWI is associated with some degree of oxidative stress [138–140], the extent of ROS production following post-exercise CWI, and the effect this has on the adaptive response to exercise, is largely unknown.

The response of post-exercise CWI on p53-related signalling has also been investigated [113, 117]. In contrast to the response to other non-genotoxic stresses, such as fluctuations in oxygen or carbohydrate availability [141, 142], CWI was reported to have no effect on $p53^{Ser15}$ phosphorylation up to 3 h following a single session of sprint-interval cycling [117]. Similarly, p53 mRNA content was unchanged when CWI was administered following a single session of sprint-interval [117] or high-intensity [113] cycling. Post-exercise CWI also has no effect on the mRNA content of p53-target genes, namely Mfn2, dynamin-1-like protein, apoptosis-inducing factor and synthesis of COX [117]. Taken together, these data suggest post-exercise CWI has no effect on p53-related signalling pathways in human skeletal muscle within 3 h post-exercise. However, given the moderate CWI-induced changes in $p53^{Ser15}$ phosphorylation and p53-target genes reported by Broatch et al. [117], and the role of p53 in mitochondrial adaptations [102, 143], further research is warranted. For example, given the paucity of research investigating the acute effects of exercise on the p53 pathway, it is possible that the 3-hour-postrecovery time point may not have

captured the maximal increase in the phosphorylation of this protein or the mRNA content of its target genes [117].

Another potential mechanism by which CWI may modify the response to endurance exercise is through the induction of CSPs. Given their role in mRNA stabilisation and the facilitation of global protein synthesis under cold stress, an up-regulation of CSPs following post-exercise CWI may serve to improve the transcription and translation of key regulatory proteins (e.g., PGC-1α) [55, 144]. Furthermore, this function may last up to 1 h post-CWI, at which time muscle temperature may still be below normothermia [145, 146]. In the only published study investigating the effects of post-exercise CWI on CSPs, CWI was reported to induce a small (non-significant) increase in RBM3 mRNA content (~ 30%), and no change in CIRP mRNA, 3 h after a single sprint-interval training (SIT) session when compared with a control condition [117]. Despite their apparent role in adaptive pathways, research investigating the role of CSPs in exercise and CWI is currently lacking and further research is warranted.

4.3 Responses to Resistance Exercise

The most notable adaptations to resistance training are an increase in strength and skeletal muscle mass. Increases in strength are generally attributed to increased skeletal muscle mass [147], enhanced neural activation [148], enhanced muscle fibre pennation angle [147], as well as possible contributions from greater muscle fibre specific tension [149] and lateral force transfer between sarcomeres and the extracellular matrix [150]. To date, the effects of CWI on the adaptations that contribute to increased strength following resistance training have not been investigated, with the exception of muscle mass.

Changes in muscle mass occur according to the balance between MPS and MPB [155]. A single resistance exercise session typically results in increased rates of post-exercise MPS that can last for up to 72 h [156, 157]. MPB also increases following resistance exercise [158]; however, this increase is suppressed when resistance exercise is performed in a state of hyperaminoacidemia [157]. Some studies have reported that post-exercise CWI reduces resistance training-induced increases in muscle mass [153, 154], suggesting that cold exposure may alter the molecular response to resistance exercise. The potential mechanisms by which cold exposure may influence the molecular response to resistance exercise are outlined below. A summary of studies investigating the effects of cold exposure following resistance exercise is presented in Table 2.

Resistance-exercise-induced increases in MPS are primarily due to activation of the mammalian target of rapamycin complex 1 (mTORC1) signal transduction pathway, which, via its downstream targets ribosomal protein S6 kinase beta-1 (p70S6 K) and eukaryotic translation initiation factor 4E-binding protein 1 (4E-BP1), controls the efficiency of protein translation [159–161]. Protein translation depends not only on translation efficiency but also the capacity for protein translation, which is dependent on ribosomal content. Accordingly, ribosomal biogenesis is likely to be important for muscle hypertrophy, as demonstrated during synergist ablation-induced overload in rodents [159]. Recent evidence suggests it may also play a role in regulating protein synthesis rates following resistance exercise in humans [162, 163].

The myostatin-Smad signalling pathway is a negative regulator of skeletal muscle growth [164, 165]. Myostatin-Smad signalling is depressed following a single resistance exercise session, as shown by increased mRNA expression of a myostatin target gene, mighty [166]. The increase in mighty mRNA following a single session of resistance exercise was very closely related to the increase in muscle mass following resistance training [166], suggesting a role for myostatin-Smad signalling in resistance exercise-induced muscle growth.

Prolonged cold exposure (> 24 h) has been shown to inhibit skeletal muscle growth in rodents, partially due to reduced protein synthesis [52, 53]. Cold-induced attenuation of protein synthesis may occur via several mechanisms. Cold reduces skeletal muscle blood flow [167–169], which is positively associated with rates of MPS [170, 171]. Thus, cold-induced vasoconstriction may reduce protein synthesis. Cold also suppresses protein translation, partially via inhibition of mTOR signalling [153, 172], possibly due to cold-induced activation of the mTOR repressor, AMPK [52, 107]. Cold exposure may also reduce protein synthesis due to inhibition of ribosomal biogenesis [152]. Increases in post-exercise protein synthesis appear to be partially dependent on the short-term exercise-induced inflammatory response [173], which may or may not be attenuated by cold exposure [151, 174].

Cold exposure also has the potential to stimulate MPS. Cold increases myoplasmic free Ca^{2+} [65], which can activate mTOR signalling [72], although it may also cause calcium-dependent proteolysis. Circulating catecholamine concentrations are increased during or following cold exposure [52, 75] and are known to stimulate protein synthesis [175]. Cortisol inhibits protein synthesis [176], and this hormone appears to be decreased following cold exposure [177]. The cold-shock protein RBM3 is induced in skeletal muscle following cold exposure [178] and promotes protein synthesis [55], as well as inhibiting apoptosis [178]. Thus, cold exposure has the potential to inhibit or stimulate rates of MPS.

Roberts et al. [153] found that CWI reduced the resistance-exercise-induced activation of p70S6 K in human

Table 2 A summary of current evidence regarding the short- and long-term effects of cold exposure following resistance exercise on skeletal muscle responses

Study	Sample size (sex)	Species	Exercise protocol	Cold intervention		Muscle alterations (vs control)	
				Single	Repeated	mRNA	Protein/other
Peake et al. [151]	9 (M)	Human	Lower-body Rx exercise	CWI ~ 10 °C for 10 min		↔ MAC1, CD163 ↔ cytokines ↔ chemokines ↔ neurotrophins	↔ neutrophil infiltration ↔ macrophage infiltration ↔ HSP70 ↔ αB-crystallin
Figueiredo et al. [152]	9 (M)	Human	Lower-body Rx exercise	CWI ~ 10 °C for 10 min		↓ pre rRNA expression ↓ rDNA transcription gene expression	↓ p-p38$^{Thr180/Tyr182}$ ↓ p-MNK1^{Thr197} ↓ p-eIF4E^{Ser209} ↓ eIF4E ↓ Cyclin D1 ↓ p-Akt^{Thr308} ↓ p-PRAS40^{Thr246} ↓ p38-MNK1-eIF4E axis signalling ↓ Akt signalling ↓ rDNA transcription signalling
Roberts et al. [153]	9 (M)	Human	Single-leg Rx exercise	CWI ~ 10 °C for 10 min			↓ total p70S6 K ↓ p-p70S6 K$^{Thr421/Ser424}$ ↓ p-p70S6K^{Thr389} ↔ p-4E-BP1 ↓ total rpS6 ↓ satellite cells
	21 (M)	Human	Lower-body Rx training 2 day/week, 12 week		CWI ~10 °C for 10 min		↓ type II fibre CSA ↓ myonuclei per fibre ↓ muscle mass
Yamane et al. [154]	14 (M)	Human	Handgrip exercise, 5 sets × 8 reps (70–80% 1RM) 3 day/week, 6 week		CWI Lower arm 10 °C for 20 min		↓ forearm flexor muscle thickness
Yamane et al. [31]	16 (M)	Human	Handgrip exercise, 3 sets × 8 reps (70–80% 1RM) 3 day/week, 4 week		CWI Lower arm 10 °C for 20 min		↓ forearm flexor muscle thickness

Akt protein kinase B, *CD163* cluster of differentiation 163, *CSA* cross-sectional area, *CWI* cold-water immersion, *eIF4E* eukaryotic translation initiation factor 4E, *HSP70* heat-shock protein 70, *MAC1* macrophage-1 antigen, *MNK1* mitogen-activated protein kinase–interacting serine/threonine-protein kinase 1, *p38* mitogen-activated protein kinase, *PRAS40* protein-rich AKT1 substrate 1, *p70S6K* ribosomal protein S6 kinase beta-1, *rDNA* ribosomal deoxyribonucleic, *rpS6* ribosomal protein S6, *rRNA* ribosomal ribonucleic acid, *Rx* resistance, *1RM* 1-repetition maximum, *4E-BP1* eukaryotic translation initiation factor 4E-binding protein 1, ↑ significant ($p < 0.05$) increase, ↓ significant ($p < 0.05$) decrease, ↔ no effect ($p > 0.05$

skeletal muscle, which suggests CWI may attenuate post-exercise MPS. In contrast, CWI caused greater phosphorylation of extracellular signal-regulated kinases 1 and 2 (ERK1/2). However, CWI had no effect on the activation of the downstream target of p70S6 K and ERK1/2, ribosomal protein S6, or other kinases involved in the

regulation of protein synthesis, such as 4E-BP1 [153]. CWI following a single resistance exercise session inhibited several key markers of ribosomal biogenesis; however, the content of mature ribosomal RNA (rRNA) components was unchanged [152]. This latter finding is not surprising given that rRNA appears unresponsive to a single session of resistance exercise [162]; multiple exercise sessions may be needed to determine the effects of CWI on rRNA content. Overall, there is weak evidence to suggest that CWI inhibits key molecular events involved in the regulation of MPS following resistance exercise. Importantly, whether CWI affects the rate of MPS after resistance exercise has not been investigated.

MPB occurs via three proteolytic systems; the autophagic-lysosomal system, apoptosis, and the ubiquitin proteasome pathway (UPP). The UPP is estimated to be responsible for 80–90% of intracellular protein turnover [179], and has been the most studied proteolytic pathway following resistance exercise. Several studies have demonstrated that key markers of UPP are responsive to both single and repeated resistance exercise sessions in human skeletal muscle at both the gene and protein level [180–182], indicating that UPP is likely to be involved in regulating muscle protein turnover following resistance exercise. It is important to recognise that MPB is a critical part of skeletal muscle remodelling. Thus, increased MPB may not necessarily be detrimental, especially with regards to damaged or unwanted proteins [155].

Prolonged cold exposure (> 24 h) has been shown to inhibit skeletal muscle growth via increased rates of calcium-dependent and ubiquitin–proteasome mediated proteolysis in rat skeletal muscle [52, 53]. Increased proteolysis was also associated with increased atrogin-1 and muscle RING-finger protein-1 (MuRF1) protein content, decreased Akt and forkhead-box class O (FoxO) phosphorylation, and increased AMPK phosphorylation [52]. Calcium-dependent proteolysis may also occur via cold-induced increases in intracellular [Ca^{2+}] [65]. Rewarming following cold exposure may also contribute to protein breakdown, as it can induce oxidative stress and result in increased apoptosis and autophagy [50]. In contrast, cold may also inhibit protein breakdown. For example, post-exercise cold exposure in humans was shown to increase PGC-1α mRNA [51, 64], an inhibitor of FoxO3 [183], which regulates pathways involved in apoptosis, autophagy, and UPP. Also, cold increases circulating catecholamines [52, 75], which inhibit proteolysis [175].

Resistance exercise increases rates of MPB when performed in a fasted state, and activates protein breakdown via the UPP. To date, no studies have investigated whether CWI alters rates of MPB or activation of UPP following resistance exercise. As mentioned above, cold exposure has the potential to stimulate or inhibit each of these so this is an important omission that warrants investigation.

Satellite cells are essential for muscle regeneration following injury; however, their role in resistance exercise-induced muscle hypertrophy is currently under debate. For example, satellite cell depletion had no effect on murine muscle growth during short-term synergist ablation [184], but did attenuate hypertrophy during more prolonged overload [185]. Using more physiologically relevant prolonged resistance exercise models in humans, individuals with the greatest pre-training satellite cell population achieved the highest muscle growth [186], and muscle hypertrophy was matched by increased satellite cell content [187]. Thus, growing evidence suggests satellite cells do play a role in resistance-exercise-induced muscle hypertrophy.

There is currently scant evidence regarding the effects of cold exposure on satellite cell content or activity. CWI blocked the increase in satellite cell content following resistance exercise in humans [153]. Similarly, ice application following muscle injury retarded the proliferation and differentiation of satellite cells [188]. Reduced differentiation following cold exposure may be due to reduced expression of myogenin [189], which is involved in terminal differentiation of myoblasts [190]. Conversely, cold exposure activates the transient receptor potential A1 channel [191], which has been shown to play a role in myoblast migration and fusion [192].

To date, only one study has investigated the effects of CWI on satellite cell response to resistance exercise. Roberts et al. [153] used two independent markers of satellite cell content, paired box protein (Pax7) and neural cell adhesion molecule (NCAM). CWI completely blocked the resistance-exercise-induced increase in $Pax7^{+}$ satellite cells up to 48 h after exercise, and delayed the appearance of $NCAM^{+}$ satellite cells [153]. Thus, CWI appears to block or delay the proliferation of satellite cells following resistance exercise, which may result in a reduced hypertrophic response.

There appear to be many potential mechanisms by which CWI may alter the molecular responses to resistance exercise and thereby potentially inhibit or promote training adaptations. Many of these mechanisms are speculative and have not been verified experimentally, thus considerable further research is needed.

5 CWI and Adaptations to Exercise Training

As highlighted in Sect. 4, CWI may alter several cellular responses associated with the adaptive response to a single exercise session. However, while the cumulative effects of transient changes in molecular pathways after each exercise session are believed to contribute to subsequent training adaptations, limited research has investigated the

effects of regular post-exercise CWI on adaptations to repeated exercise sessions (i.e., exercise training). This section will focus on the potential influence of regular post-exercise CWI on long-term adaptations to exercise training.

5.1 Endurance Exercise Training

A number of studies have investigated the merit of regular post-exercise CWI during an endurance training period (i.e., 3–6 weeks) (Fig. 3). An attenuation of endurance performance and maximal oxygen uptake was observed following 4–6 weeks of regular post-exercise CWI (5 °C for 2 × 20 min), as compared with a passive control [31]. In contrast, subsequent research employing more commonly utilised immersion protocols (e.g., 10–15 °C CWI for 10–15 min) has reported minimal effects of regular CWI on endurance training adaptations [41, 117, 119]. For example, regular CWI during 3–4 weeks of intense cycling training was shown to have negligible [117, 119] or slightly beneficial [41] effects on cycling performance. In support of this, regular CWI following sprint-interval cycling had no effect on the lactate threshold, maximal oxygen uptake, and peak aerobic power [117]. However, conclusive evidence either supporting or refuting the merit of CWI on adaptations to endurance exercise training is lacking. As such, a number of studies have investigated the effects of post-exercise CWI on molecular markers of endurance-related adaptations in human skeletal muscle.

Three studies have investigated the effects of regular post-exercise CWI on molecular markers of endurance training adaptations in human skeletal muscle, with mixed results. Although the post-exercise increase in PGC-1α mRNA content has been reported to be augmented following CWI [111–113], evidence for a subsequent increase in protein content is limited. A single post-exercise CWI exposure does not appear to provide a sufficient stimulus to increase PGC-1α protein content, with two studies reporting no change in content up to 3 h after exercise [112, 113]. Similarly, regular post-exercise CWI during 6 weeks of cycling SIT [117], or 4 weeks of running high-intensity interval training (HIT) [120], had no effect on PGC-1α protein content. Continued work in this area, including potential CWI-induced alterations in PGC-1α post-translational modification and nuclear translocation, will help clarify the above-mentioned inconsistencies.

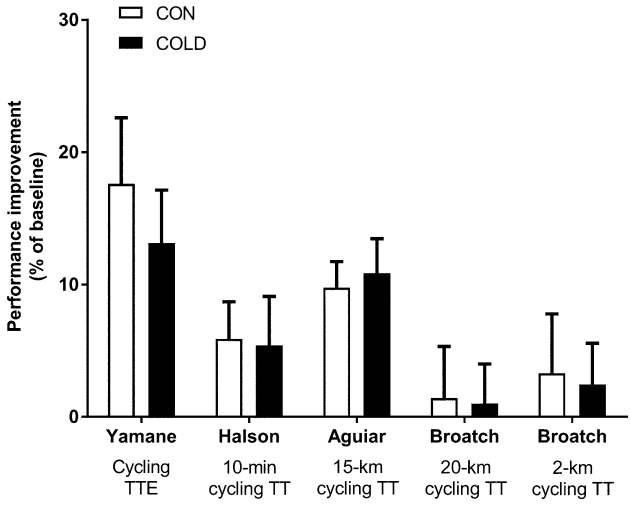

Fig. 3 The effects of regular post-exercise cold-water immersion (*COLD*) on markers of endurance exercise performance following 3–6 weeks of cycling training. *CON* control, *TTE* time to exhaustion, *TT* time trial. Data is from studies by Yamane et al. [31], Halson et al. [41], Aguiar et al. [119] and Broatch et al. [117]

The effects of regular post-exercise CWI on other markers of mitochondrial adaptations have also been investigated. Ihsan et al. [120] reported significant increases in β-HAD protein, and OXPHOS complexes I and III content, yet no change in PPARα, and OXPHOS complexes II, IV and V content following 4 weeks of cycling HIT and regular CWI. In the only study to investigate the effects of regular post-exercise CWI on p53-related adaptations, no change in p53 content was observed following 6 weeks of cycling SIT and regular CWI [117]. Markers of mitochondrial function, including respiration and enzyme activities (CS, carnitine palmitoyltransferase I and succinate dehydrogenase) are also unaltered with regular post-exercise CWI [117, 119]. The effects of regular post-exercise CWI on mitochondrial content have been assessed via CS activity [117]; however, no study has assessed this using transmission electron microscopy (TEM) imaging, the gold standard for measuring mitochondrial content [193]. Taken together, conclusive evidence for CWI-induced increases in the content of proteins associated with mitochondrial biogenesis, or improvements in mitochondrial function and/or content, in human skeletal muscle is lacking.

Considering the reported increases in mitochondrial proteins in animals exposed to cold air for 24 h per day [65, 107, 109, 126], a larger cold-stress (i.e., colder and/or longer) may be needed to elicit significant changes in human skeletal muscle. For example, 9 weeks of cold acclimation (4–18 °C for 8–9 weeks) has been reported to increase PGC-1α protein content by ~ 1.5-fold in mice [65]. Prolonged cold exposure has also been reported to improve other markers of endurance-training adaptations, including increases in slow-twitch muscle fibre ratio, basal calcium levels, and mitochondrial enzyme activity (e.g., CS and COX) [65, 109, 126, 194]. Further work is needed to elucidate if these responses are conserved in humans.

5.2 Resistance Exercise Training

Several studies have investigated the effects of post-exercise CWI on adaptations to resistance training, with mixed

results (Fig. 4). Regular post-exercise CWI performed during resistance training has been shown to attenuate increases in muscle strength [153, 154], endurance [31] and mass [153, 154] within the cooled limb. Conversely, in other studies CWI had no detrimental effects on muscle strength [31, 195, 196], endurance [31, 154, 196] or mass [31, 196]. The reasons for the conflicting findings are difficult to ascertain, as studies have varied markedly with respect to study design (within or between subjects), CWI temperature and duration, and resistance training programme. Even studies with identical immersion protocols and very similar training programmes have reported conflicting effects of CWI [31, 154, 196].

Research investigating the effects of regular post-exercise CWI on molecular markers of skeletal muscle adaptations to resistance training in human skeletal muscle is limited. Consistent with reported reductions in muscle mass and type II muscle fibre cross-sectional area, Roberts et al. [153] also reported regular post-exercise CWI during resistance training to reduce the content of myonuclei per muscle fibre. However, research investigating the effects of regular post-exercise CWI on molecular regulators of MPS (e.g., ribosomal content and mTOR content), and MPB (e.g., MuRF1, atrogin-1 and FoxO content), is lacking.

6 Generic Considerations and Limitations

Limitations exist in many studies that limit their applicability to an athletic population. For example, some studies have immersed only a single limb [31, 111, 113, 120, 154, 195, 196], and/or used training programmes involving only a single exercise [31, 154, 195, 196] or lower-body-only exercises [153]. No study has investigated the effects of CWI on adaptations to whole-body resistance training. Since the molecular responses to whole-body resistance training differ to lower-body-only training [197], it is possible that the effects of CWI may also differ between whole-body and lower-body-only resistance exercise.

An additional limitation is that the training frequency and volume used in previous studies is less than that normally performed by athletes. Athletes may train between 5 and 14 times per week [198], with limited time for recovery between training sessions. If CWI can hasten recovery then it may improve training performance and presumably the stimulus for adaptation. In contrast, studies to date have used training frequencies of 2 to 3 times per week; this may allow adequate recovery between sessions regardless of the recovery-enhancing benefits of CWI. It remains to be determined whether CWI alters training adaptations in situations where recovery between training sessions may be limited.

Research is also required to further investigate alternate mechanisms by which CWI may alter the adaptive response to exercise. These include cold-induced oxidative stress, cold/heat-shock proteins, and alternate signalling pathways (e.g., stress-activated protein kinases, CaMK, autophagic-lysosomal breakdown, and apoptosis). Novel techniques such as TEM for the assessment of mitochondrial content, and methods to assess rates of MPS, will also contribute to understanding the mechanisms by which post-exercise CWI may alter the adaptive response to exercise. Additionally, investigations on the optimal 'dose' of exercise and post-exercise CWI to enhance human skeletal muscle adaptations to exercise are warranted.

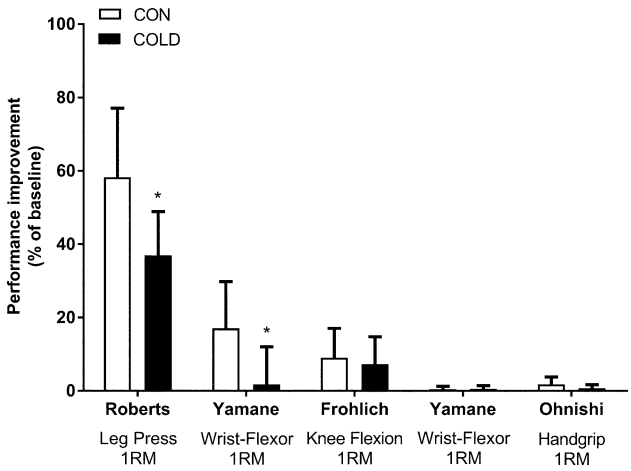

Fig. 4 The effects of regular post-exercise cold-water immersion (*COLD*) on 1-repetition maximum (*1RM*) strength following resistance training. *CON* control. * Significantly different to *COLD*. Data is from studies by Roberts et al. [153], Yamane et al. [154], Frohlich et al. [195], Yamane et al. [31] and Ohnishi et al. [196]

7 Conclusion

Evidence for a positive effect of post-exercise CWI on endurance adaptations is scarce. Despite reported short-term increases in markers of mitochondrial biogenesis following a single session of post-exercise CWI (e.g., PGC-1α mRNA and p-AMPK), regular post-exercise CWI appears to have little to no effect on long-term adaptations. Considering the well established effects of cold on mitochondrial adaptations in animal models (Table 1), it is possible that the CWI protocols used to date do not provide a large-enough cold stimulus, and longer and/or colder immersion may be needed to elicit significant alterations in mitochondrial adaptations in human skeletal muscle. However, apart from early work by Yamane et al. [31], it is important to note that no research has reported a negative effect of post-exercise CWI on markers of adaptation to endurance exercise or exercise performance. As such, current evidence suggests no reason to incorporate CWI

into an endurance training programme if the goal is to augment exercise-induced mitochondrial adaptations.

It is also difficult to make clear conclusions and recommendations regarding the effects of CWI on adaptations to resistance training. Studies have reported either no effect or a negative effect of CWI on key resistance training adaptations such as strength and muscle mass. Likewise, evidence to date suggests either no effect or negative effects of CWI on molecular mechanisms involved in regulating adaptations to resistance training. Importantly, no study has shown beneficial effects of post-exercise CWI on resistance molecular responses or training adaptations. However, caution is needed in interpreting these studies as they may not apply to all training scenarios. Nonetheless, based on the evidence to date, there seems no rationale for incorporating CWI into a resistance training programme. Whether this is true during periods of high-frequency training, with limited recovery between sessions, remains to be determined.

Compliance with Ethical Standards

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